CAMPUS HELP BOT

Rule-Based Python Chatbot for Student Support

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TASK OBJECTIVE

Build a simple conversational chatbot in Python that asks for the student's name, runs continuously, recognizes greetings, answers common campus-related questions, handles unknown messages, and exits when the user enters **bye** or **exit**.

TIME PLAN

Time	Task
0-5 min	Plan the chatbot and decide the student-support topics.
5-20 min	Write and test the Python chatbot.
20-25 min	Add extra responses and test different inputs.
25-30 min	Clean the code and prepare a quick demonstration.

REQUIREMENTS

No.	Requirement
1	Ask the user's name before the main conversation.
2	Use while True: to keep the conversation running.
3	Use input() to receive messages.
4	Clean the input using .lower().strip().
5	Recognize at least two greetings, such as hello and hi.
6	Answer at least three different campus-related topics.
7	Use if / elif / else for decision making.
8	Use break to exit when the user types bye or exit.
9	Use the user's name in at least one response.
10	Provide a useful response for unknown messages using else.
11	Use the in operator for flexible greetings such as hello bot.

Challenge: Use the **in** operator so variations such as “hello bot” or “hey hello” can still trigger a greeting.

Theme: Student support - campus facilities, library, timetable, study tips, and general guidance.

1. PYTHON CODE

Complete runnable code for the Campus Help Bot.

```
# Campus Help Bot - Rule-Based Python Chatbot

print("=" * 52)
print("                CAMPUS HELP BOT")
print("            Student Support Assistant")
print("=" * 52)

name = input("Enter your name: ").strip()

print(f"\nWelcome, {name}!")
print("Ask me about:")
print("- Library")
print("- Timetable")
print("- Campus facilities")
print("- Study tips")
print("Type 'bye' or 'exit' to stop.\n")

while True:
    message = input(f"{name}: ").strip().lower()

    if message == "bye" or message == "exit":
        print(f"Bot: Goodbye, {name}! Have a great day!")
        break

    elif "hello" in message or "hi" in message or "hey" in message:
        print(f"Bot: Hello {name}! How can I help you today?")

    elif "library" in message or "book" in message:
        print("Bot: The library is the place to borrow books, use study areas, "
              "and access academic resources.")

    elif "timetable" in message or "schedule" in message:
        print("Bot: Check your department timetable for class timings, "
              "rooms, and practical sessions.")

    elif "facility" in message or "canteen" in message:
        print("Bot: Campus facilities may include the canteen, labs, "
              "library, sports areas, and student support services.")

    elif "study" in message or "exam" in message:
        print("Bot: Make a daily study plan, revise important concepts, "
              "practise questions, and take short breaks.")

    else:
        print("Bot: I can help with library, timetable, campus facilities, "
              "or study tips.")
```

SAMPLE EXECUTION OUTPUT

```
==================================================
                CAMPUS HELP BOT
            Student Support Assistant
==================================================
Enter your name: M.Rehana

Welcome, M.Rehana!
Ask me about:
- Library
- Timetable
- Campus facilities
- Study tips
Type 'bye' or 'exit' to stop.

M.Rehana: hello bot
Bot: Hello M.Rehana! How can I help you today?

M.Rehana: study tips
Bot: Make a daily study plan, revise important concepts,
practise questions, and take short breaks.

M.Rehana: bye
Bot: Goodbye, M.Rehana! Have a great day!
```

The sample output demonstrates name input, flexible greeting detection, a topic response, and the exit condition.

2. RESULT

The test cases below cover the required chatbot behaviours.

Test Input	Expected Result	Observed Result	Status
hello	Greeting response	Greeting displayed	PASS
HELLO	Case-insensitive greeting	Greeting displayed	PASS
hello bot	Flexible greeting using in	Greeting displayed	PASS
hi	Greeting response	Greeting displayed	PASS
LIBRARY	Library guidance	Library guidance displayed	PASS
TIMETABLE	Timetable guidance	Timetable guidance displayed	PASS
CANTEEN	Campus facilities response	Facilities guidance displayed	PASS
STUDY TIPS	Study response	Study advice displayed	PASS
random message	Useful fallback using else	Fallback displayed	PASS
bye	Exit message and break	Chatbot stopped	PASS

PROJECT FLOW

INPUT	PROCESS	RESPOND	REPEAT	EXIT
User enters a message	lower().strip() + if/elif/else	Bot selects a suitable reply	while True keeps chat active	bye / exit + break

FINAL VERIFICATION

The Campus Help Bot demonstrates the required Python concepts and the complete conversation flow: **INPUT** → **PROCESS** → **RESPOND** → **REPEAT** → **EXIT**. It provides useful student-support responses while remaining simple, readable, and runnable.

Mini Project: Campus Help Bot

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